Universidad Europea

RETAIL FLOW

Group: Business I

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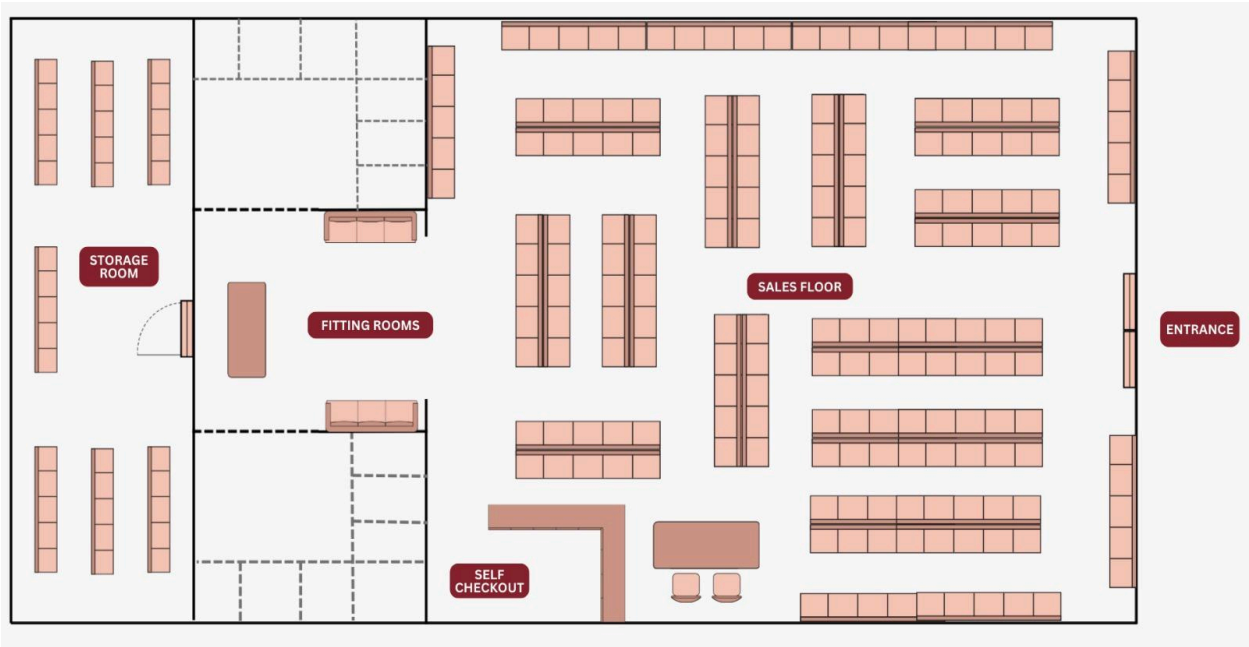
1. Spatial Infrastructure

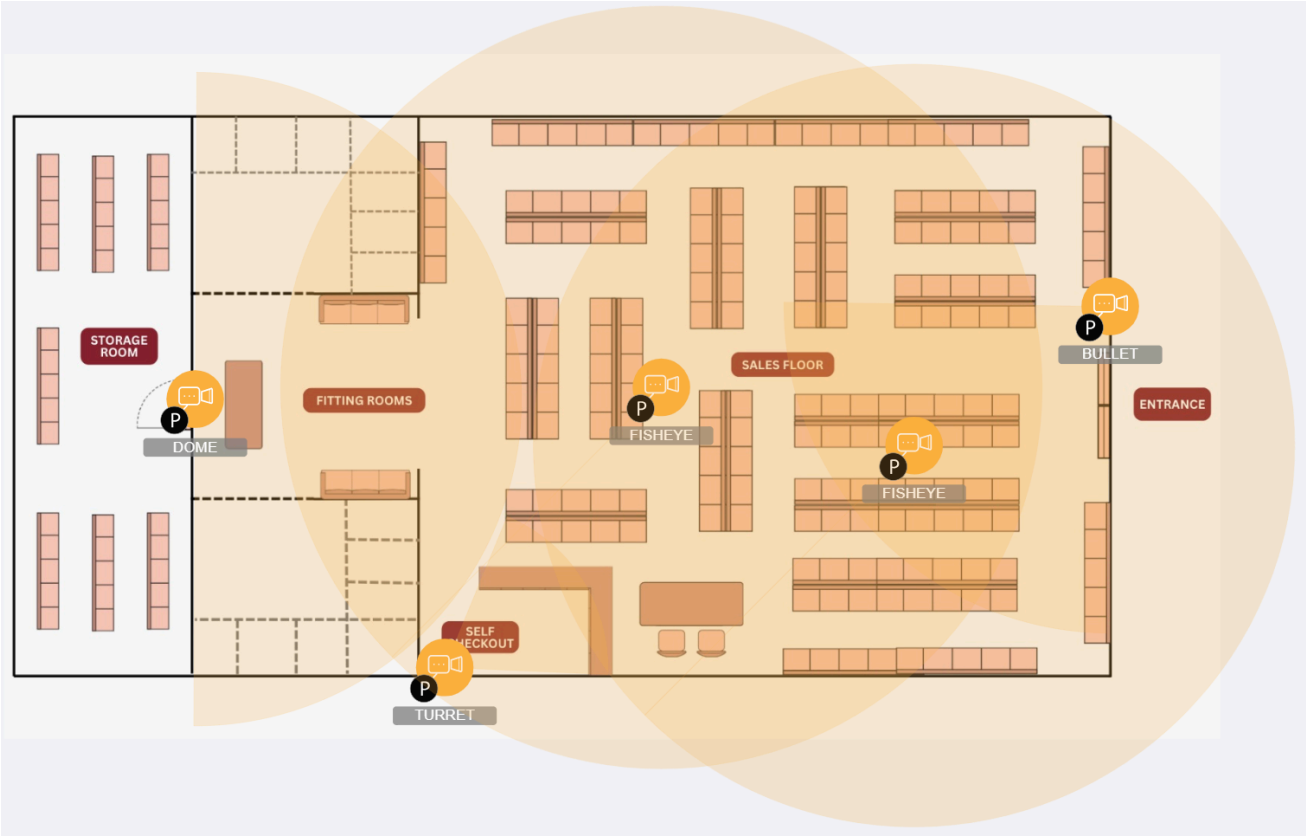
The store operates within a defined 100 ft x 50 ft surface area that is divided into five key functional zones. This layout is designed to facilitate a natural flow, guiding customers through high-engagement areas before reaching the final transaction point.

- **Entrance:** A singular, controlled entry and exit point that will be used to count people going in and out.
- **Sales floor:** This is where we can find the shelving units with the products, making it our primary data source.
- **Fitting rooms:** A small lobby consisting of a customer lounge with sofas and fitting rooms that will help us analyze the products that are capturing the customers' attention.
- **Self-checkout:** This is the final point of the customer where we can capture data focused on the transactions and see what's selling more.
- **Storage:** A restricted area for inventory management.



2. Technical Infrastructure

Our system employs an advanced array of five specialized cameras to convert physical movement into a digital data stream. By implementing targeted lens technologies in critical areas, we guarantee precise capture for both security and behavioral analysis.



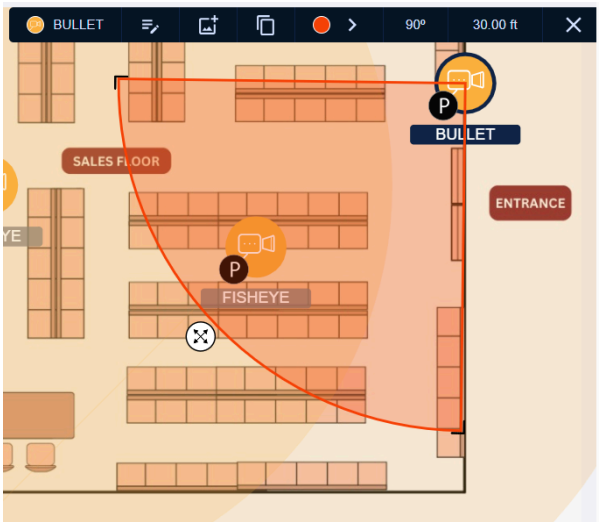
- **Camera 1: Entry/Exit Control**

  Camera Type: Fixed-Lens Bullet

  Viewing range: 90º angle & 30 ft radius

  Placement: Mounted above the primary entrance, oriented toward the interior.

  Purpose: Records entry/exit events to calculate store-wide conversion rates.



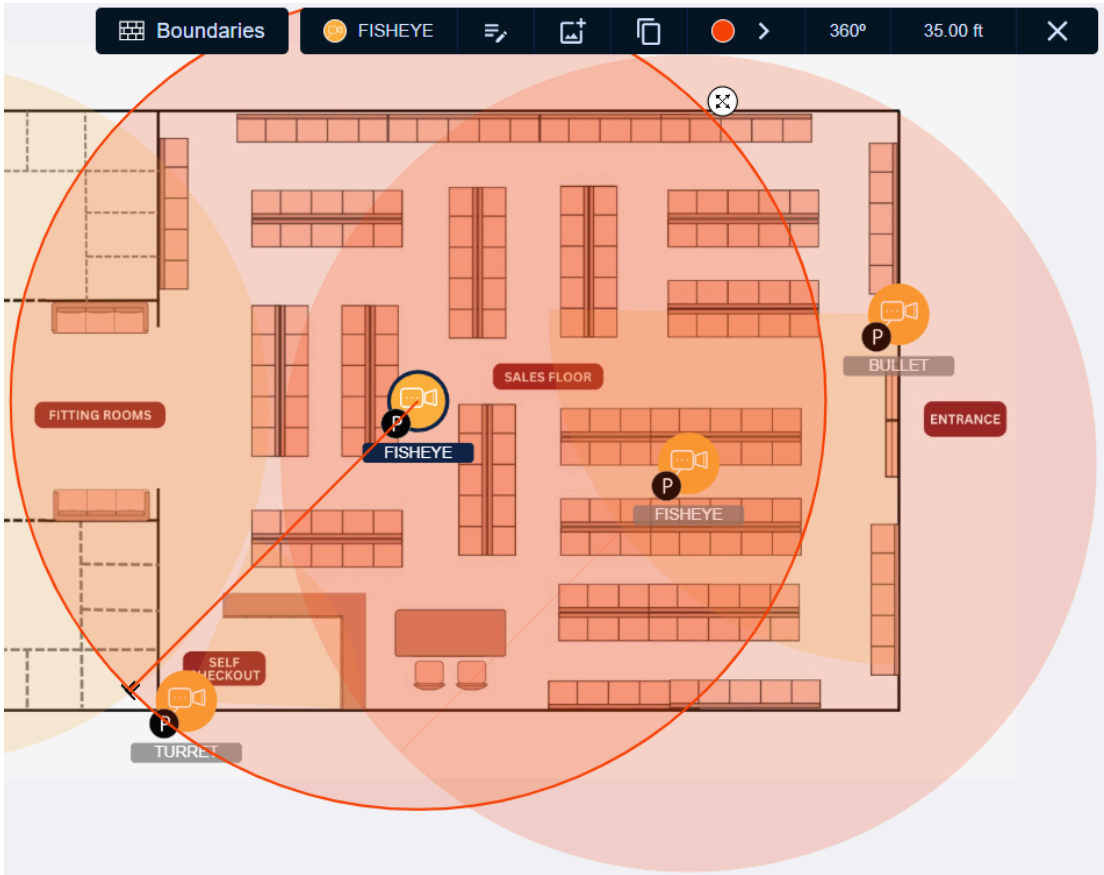
- **Camera 2 & 3: Behavioral Core**

  Camera Type: Multi-Lens Fisheye

  Viewing range: 360º angle & 35 ft radius

  Placement: Mounted in the main sales floor area with an overlap to eliminate shelf shadows.

  Purpose: Facilitates seamless path tracking and heat mapping as customers navigate the store and identify popular and ignored merchandise.



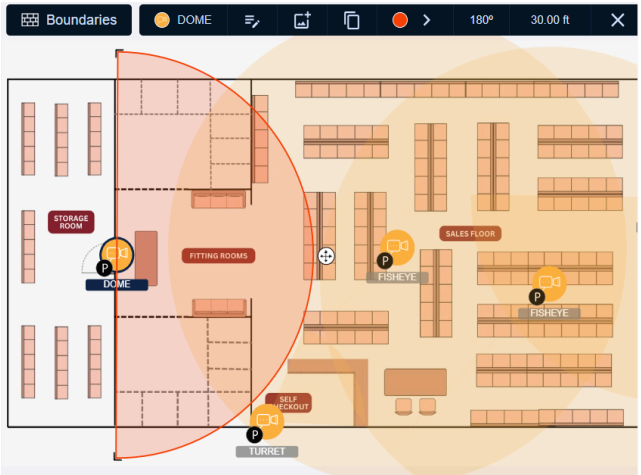
- **Camera 4: Fitting Room Monitor**

  Camera Type: Fixed Dome

  Viewing range: 90º angle & 30 ft radius

  Placement: Mounted in fitting room lobby.

  Purpose: Measures dwell time in the lounge and possible transitions from browsing to trying on.



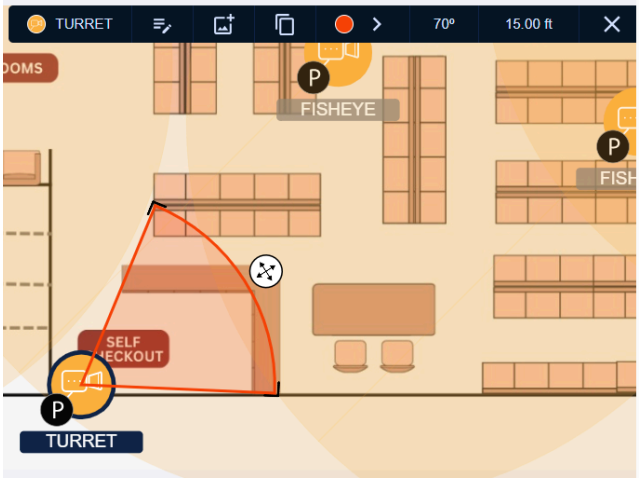
- **Camera 5: Transaction Monitor**

  Camera Type: Fixed-Lens Turret

  Viewing range: 70º angle & 15 ft radius

  Placement: Mounted directly over the self-checkout area.

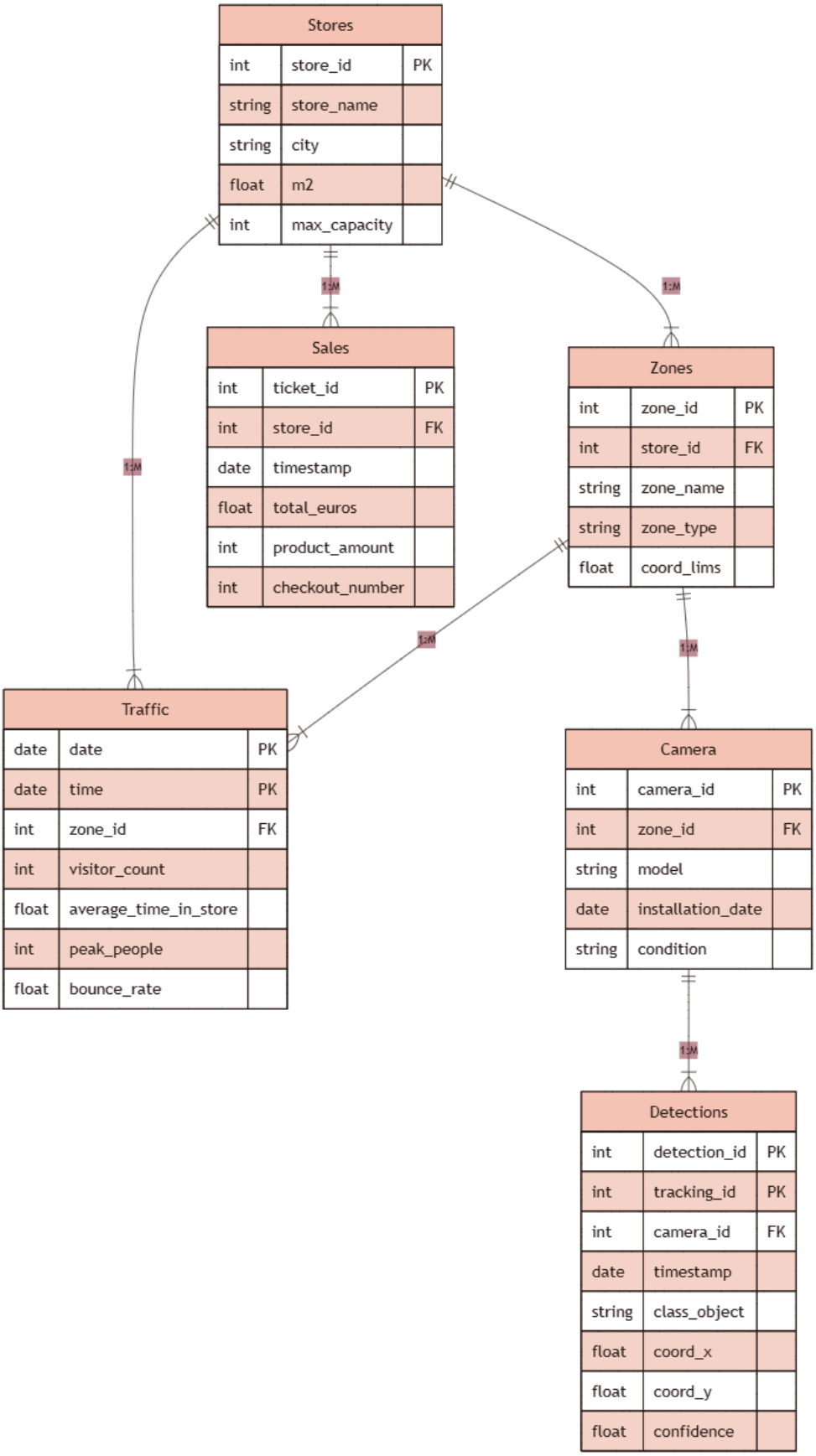
  Purpose: Correlates physical customer behavior with digital transaction logs to measure the "Popularity vs. Purchase" index.



3. Data Structure

The database is structured according to a relational model (SQL). The relationships among entities are rigorously defined through Primary Keys (PK) and Foreign Keys (FK), indicating that the creation of a "Sale" or a "Zone" is contingent upon the prior existence of a "Store".

- **Top Level:** Stores (an independent entity).
- **Level 2** (Dependencies): Zones and sales are contingent upon the stores table (one store may encompass multiple zones/sales).
- **Level 3** (Hardware): Cameras are dependent on zones (one zone may contain multiple cameras).
- **Level 4** (Data Stream): Detections are reliant on the camera (one camera can generate numerous detections).
- **Level 5** (Analytics): Traffic is associated with zones (one zone may experience significant traffic).


Stores
int store_id PK
string store_name
string city
float m2
int max_capacity
1:M
1:M
Sales
int ticket_id PK
int store_id FK
date timestamp
float total_euros
int product_amount
int checkout_number
Zones
int zone_id PK
int store_id FK
string zone_name
string zone_type
float coord_lims
1:M
1:M
1:M
Traffic
date date PK
date time PK
int zone_id FK
int visitor_count
float average_time_in_store
int peak_people
float bounce_rate
Camera
int camera_id PK
int zone_id FK
string model
date installation_date
string condition
1:M
Detections
int detection_id PK
int tracking_id PK
int camera_id FK
date timestamp
string class_object
float coord_x
float coord_y
float confidence

4. Variable explanation

Variables			
Stores			
Name	**Key**	**Type**	**Real world description**
store_id	PK	int	Unique identifier for each physical store location.
store_name		string	Commercial name of the branch (e.g., "Zara Gran Via").
city		string	City where the store is.
m2		float	Total surface area of the store in square meters.
max_capacity		int	Legal maximum occupancy allowed.
Sales			
ticket_id	PK	int	Unique identifier for the transaction (receipt).
store_id	FK	int	Foreign Key linking the sale to a specific Store.
timestamp		date	Exact date and time the transaction was completed.
total_euros		float	Total monetary value of the purchase (Revenue).
product_amount		int	Total quantity of individual items in the basket.
checkout_number		int	Identifier of the specific POS/Register used.
Zones			
zone_id	PK	int	Unique identifier for a logical area within the store.
store_id	FK	int	Foreign Key indicating which Store this zone belongs to.
zone_name		string	Descriptive name (e.g., "Fitting Rooms", "Entrance").
zone_type		string	Category of the area (Sales, Transit, Service, Staff).
coord_lims		float	Spatial coordinates defining the zone boundaries on the map.
Camera			
camera_id	PK	int	Unique hardware identifier for the device.
zone_id	FK	int	Foreign Key indicating which Zone the camera monitors.
model		string	Hardware model or manufacturer of the sensor.
installation_date		date	Date when the device was activated.
condition		string	Operational status (Active, Maintenance, Error).
Detections			
detection_id	PK	int	Unique ID for every single detection event.
tracking_id	PK	int	Anonymized Hash representing a unique moving object.
camera_id	FK	int	The specific camera that generated this detection.
timestamp		date	High-precision timestamp (ms) of the detection.
class_object		string	Classification of the object (Person, Staff, Cart).
coord_x		float	Normalized X-coordinate position on the floor plan.
coord_y		float	Normalized Y-coordinate position on the floor plan.
confidence		float	AI Probability score (0.0 - 1.0) of the classification.
Traffic			
date	PK	date	Date of the record (YYYY-MM-DD).
time	PK	date	Hour slot of the record (0-23).
zone_id	FK	int	The specific Zone being analyzed.
visitor_count		int	Total number of unique tracking_ids detected.
average_time_in_store		float	Average dwell time (seconds) per visitor.
peak_people		int	Maximum number of simultaneous people detected.
bounce_rate		float	Percentage of visitors who leave after entering the store without interacting.

5. Data source

In the context of our project, we did not need to obtain any public data; all the information will be sourced directly from the video footage captured by the cameras in our stores.

- Sales: Data in the sales table originates from the physical cash registers.
- Detections & traffic: The data represented in the detections and traffic table is sourced from the cameras (as detailed in the camera table) that process video feeds in real-time.
- Static configuration: Data in stores, zones, and cameras is static infrastructure data defined by the business layout.

6. Bibliography

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